

VR training for critical tasks - assembling the INSPIRE emergency pulmonary ventilator for COVID-19 treatment

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ABSTRACT

This paper describes a VR system designed to train health professionals and volunteers to properly assemble the replaceable pneumatic circuit of the Inspire emergency pulmonary ventilator. The simulator's primary goal is to internalize a mindful attitude in the trainee, as even minor mistakes may lead to disastrous consequences during the machine's operation, such as sequels and even patient death. The system was based on the state of the art of medical VR training and proposes innovative ways to communicate and demonstrate instructions, possible repercussions of mistakes, and feedback mechanisms to portray step completion and execution accuracy. The main challenge was dealing with standard VR systems' limitations, particularly the absence of force feedback.

Index Terms: Human-centered computing: Human computer interaction (HCI) - Interaction Techniques

1 INTRODUCTION

The COVID 19 pandemics deeply impacted all aspects of ordinary life. In most countries, Intensive Care Units (ICUs) were quickly occupied, and healthcare systems collapsed, unable to treat all severe cases. The high mortality rate is associated with Acute Respiratory Distress Syndrome (ARDS), demanding protective pulmonary maneuvers. ARDS treatment by mechanical ventilation is not trivial, demanding constant monitoring to avoid lung damage and death. Therefore, the most critical element at ICUs are pulmonary ventilators, and the equipment was soon depleted from the global market, resulting in soaring prices, reaching more than 50,000 USD. Several organizations worldwide engaged in projects to design and manufacture emergency ventilators.

One of those projects is Inspire from Brazil, conducted by a team of volunteers at the Polytechnic School of the University of Sao Paulo. The project was backed by several funding sources, including a 200,000 USD grant from the US Office of Naval Research (ONR), and each unit costs less than 1,000 USD. The objective was to design, prototype, and assess an open-source, low-cost emergency ventilator from off-the-shelf components. Inspire is a computer-controlled manual ventilator, composed by a high-end single board computer equipped with a touchscreen interface, laser cut aluminum structure, an electric actuator, several sensors and a pneumatic circuit that can be replaced to treat a new patient (Fig. 1e). Sensors allow monitoring relevant variables in real-time but may also enable doctors to monitor patients remotely.

This project aimed to develop a simple VR simulator to assist volunteers and health professionals to learn the 20 installation steps of Inspire's replaceable pneumatic circuit. It is of utmost importance to guarantee correct assembly, as even minor mistakes can cause lung damage and death. A conventional VR system was chosen, given

cost and availability concerns. The simulator's main contribution is internalizing a mindful attitude, and the trainee will also be able to get a first contact with the ventilator, get used to the parts, memorize procedure steps, and identify the most critical moments.

2 RELATED WORK

Training is a usual commercial application of VR, due to economic, ethical, and security concerns. A common application is the simulation of risky scenarios, such as operations on power lines [1], which involves several stages and focus on compliance with security protocols. In this case, training novice operators in real conditions may result in injuries and death, warranting investments in such simulators. Military flight promoted the development of HMD systems, and simulators for military operators are in use for many decades, providing opportunities to train complex skills more often without the associated risks and costs [6]. However, simulators are also used for industrial [3] or large scale risky assembly operations [5], training workers safely to assure product quality standards.

Healthcare has also adopted VR simulators successfully, contributing to the development of force feedback devices, usually adopted on surgical procedure simulators [2]. Surgery is a high precision and high-risk operation that requires many hours of training, and the use of cadavers and animals is being progressively banned, while practice in patients is not acceptable. Practice of medical procedures helps students and professionals to build familiarity with equipment and procedure steps, prevents contamination risks and boosts confidence [4]. Those benefits are valuable for any risky operation, including assembly of medical devices.

Therefore VR training is widely used for critical task training, especially when high costs and risks are involved. The main goal is to train critical steps or the whole procedure, focusing on precise neuromotor patterns or security rules. The approach may rely solely on VR or adopt supporting instruction media, such as images, videos, or live demonstrations. Each step is presented and the execution follows, usually recorded and evaluated by the system, given the defined metrics. After the training session, the system may present statistics and feedback.

3 SYSTEM DESCRIPTION

The system hardware is based on the Vive Pro VR system, comprising a headset, two tracked controllers, and tracking stations. The tracking area was configured as a 4 x 4 meters square. The complete training experience has four stages, where the trainee progressively learns how to interact with the system: locomotion tutorial; presentation about the Inspire project; virtual instructor tutorial; basic interaction tutorial; advanced interaction tutorial.

3.1 Narrative, Mechanics and Interactions

Aside from the tutorials and project presentation stages, the training mechanics is repetitive. For each assembly step, the user receives an instruction from the tutor, sees the demonstration of that step, and then is asked to perform, assisted by a completion and precision feedback system. During the training session, the user can teleport, open doors, pick, release and connect active parts.

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Scene and narrative The virtual environment is divided into two rooms - a reception room connected to an intensive care room. Scene design pursued portraying a realistic scenario, containing all elements expected in a hospital setting, in order to place the trainee (health care professional) in a familiar situation. The narrative was designed to integrate the whole experience into a coherent package to get the user progressively familiar with the system's features and purposes.

Project presentation Starting from the waiting room's entrance, the user is instructed to teleport to one of the chairs, where one watches a short video (1 min) about the Inspire project in a virtual TV (Fig. 1a). This step intends to describe the Inspire project, starting with the problem statement - the shortage and high costs of ventilators in Brazil - proceeding to the presentation of core members and a few details about the system's design and production.

Virtual instructor This character, called Inspiration, was designed as a robot to avoid the known uncanny valley effect and the issues usually associated with humanoid characters. The use of non-humanoid avatars can also bring a ludic element to the simulation. This instructor is intended to direct the user's attention to the current step, presents relevant information with voice or text, and points out the most critical moments, where the trainee must be extra careful. In order to call the user's attention, the robot gets closer to the next piece, starts flashing and beeping, and points to the concerned part. When the user's vision focuses on the correct part, the robot describes the step during an animated demonstration. If the user makes a mistake, the instructor beeps and caution about consequences, restarting the step in sequence.

Tutorials The basic tutorial teaches the teleport procedure and how to use controller's triggers to grab or release objects. The user is also advised to restrict walking to the assembly table perimeter. The tutorial starts at the lobby entrance (fig. 1a), then the user teleports to the main room (fig. 1b), walks around the table, and takes the ventilator parts out of a box. Such elementary steps are intended to teach basic interactions. Then the advanced tutorial is performed during the first step, presenting all application mechanics to the user.

Demonstration mechanics: Each step is described with text either on a blackboard (fig. 1c) or over the robotic avatar. After the description, an animation cycles three times, moving the part from the start to the correct final position. The connection stage of the animation is slower so the user can capture the details.

Collision mechanics: Feedback is given by a color-based scheme. Connection's status is communicated by object's color - yellow means initial contact, light green partial contact and dark green full contact (fig. 1d). Precision is indicated by the thickness of an outline effect, reinforced by controller's vibration, and both effects are proportional to the interpenetration between virtual parts (Fig. 1d) - high interpenetration means little precision. Both feedback mechanisms combined are expected to warrant correct and careful execution. An algorithm calculates how much negligence was accumulated on each step, and if interpenetration instantly surpasses 5 mm or stays between 3 and 5 mm for more than 3 seconds, the user is asked to repeat the step and be more careful.

4 RESULTS AND CONCLUSION

Considering the limitations to conduct VR assessment during the COVID-19 pandemics, the system was assessed by the development team and by invited healthcare professionals, through streamed demonstration sessions. The narrative's progressive learning concept was deemed adequate, considering that most reviewers never tested VR systems. A few asked about the teleport system and why one could not simply walk freely. Most doubts were related to the limitations of current VR systems, and after further explanations, they understood the difficulties behind trackable area real state and bare hand tracking, and found both teleport and grab interactions acceptable and straightforward.

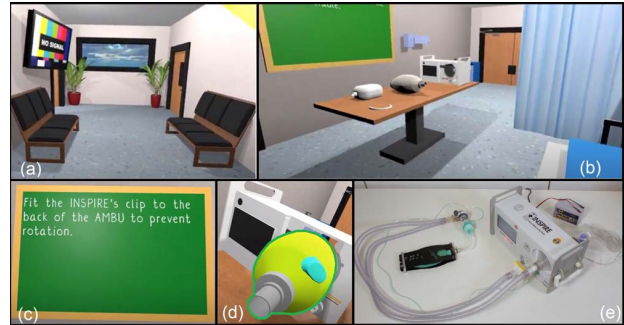


Figure 1: Waiting room and virtual TV (a). Intensive Care room and assembly table (b). Blackboard for textual instructions (c). Object's color relates to completion feedback and line thickness to precision feedback (d). Assembled Inspire emergencial ventilator (e)

Not all invited reviewers knew about the Inspire project and the presentation video was found informative. A few suggested that the video was unnecessary, since trainees would already know about the project, while others found that it could help novice users get used to the VR system. The combination of text and animation was considered a satisfactory instruction rationale, although some suggested audio explanations. However, most agreed that text might be better if the system is only available in English. The animation speed reduction at the connection stage of each step was considered an efficient way to raise trainees awareness of the important details. Finally, reviewers found that the collision feedback system induces greater care, but only after committing a few mistakes. However, the warnings on lousy execution and the possibility to repeat steps eventually teaches all users about the importance of being careful, allowing them to learn all steps correctly.

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